WHEN DOES CONTEXT HELP? A SYSTEMATIC STUDY OF TARGET-CONDITIONAL MOLECULAR PROPERTY PREDICTION

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Paper under double-blind review

ABSTRACT

We present the first systematic study of when target context helps molecular property prediction, evaluating context conditioning across 10 diverse protein families, 4 fusion architectures, data regimes spanning 67–9,409 training compounds, and both temporal and random evaluation splits. Using NESTDRUG, a FiLM-based nested-learning architecture that conditions molecular representations on target identity, we characterize both success and failure modes with three principal findings. First, fusion architecture dominates: FiLM outperforms concatenation by 24.2 percentage points and additive conditioning by 8.6 pp; how you incorporate context matters more than whether you include it. Second, context enables otherwise impossible predictions: on data-scarce CYP3A4 (67 training compounds), multi-task transfer achieves 0.686 AUC where per-target Random Forest collapses to 0.238, demonstrating that context conditioning unlocks viable prediction for novel targets where traditional approaches fail entirely. Third, context can systematically hurt: distribution mismatch causes 10.2 pp degradation on BACE1; few-shot adaptation consistently underperforms zero-shot. Beyond methodology, we expose fundamental flaws in standard benchmarking—1-nearest-neighbor Tanimoto achieves 0.991 AUC on DUD-E without any learning, and 50% of actives leak from training data, rendering absolute performance metrics meaningless. Our temporal split evaluation (train $\leq$2020, test 2021–2024) achieves stable 0.843 AUC with no degradation, providing the first rigorous evidence that context-conditional molecular representations generalize to future chemical space.

1 INTRODUCTION

Should molecular property prediction models incorporate target context? The answer is not obvious. Per-target models achieve excellent performance when training data is abundant. Multi-task models promise knowledge transfer but risk negative transfer when tasks conflict. Despite widespread interest in context-conditional architectures, *when* and *why* context helps remains poorly characterized.

We present a systematic study of this question, evaluating context conditioning across (1) 10 diverse protein families, (2) 4 fusion architectures, (3) varying data regimes (67–9,409 training compounds), and (4) temporal vs. random splits. Using NESTDRUG, a FiLM-based architecture (Perez et al., 2018) that conditions molecular representations on target identity, we map both success and failure modes of context conditioning.

Key findings. (1) *Context fusion matters*: FiLM outperforms concatenation by 24.2 pp and additive conditioning by 8.6 pp. (2) *Context helps most targets*: 9/10 targets improve (mean +5.7 pp, $p < 0.01$). (3) *Context enables data-scarce targets*: On CYP3A4 (67 actives), per-target RF collapses to 0.238 AUC while multi-task transfer achieves 0.686. (4) *Context can hurt*: BACE1 shows -10.2 pp due to distribution mismatch.

Benchmark critique. We document severe DUD-E limitations: 1-NN Tanimoto achieves 0.991 AUC without learning; 50% of actives overlap with ChEMBL training. Our temporal split (train $\leq$2020, test 2021–2024) provides rigorous evaluation, showing stable 0.843 AUC across years.

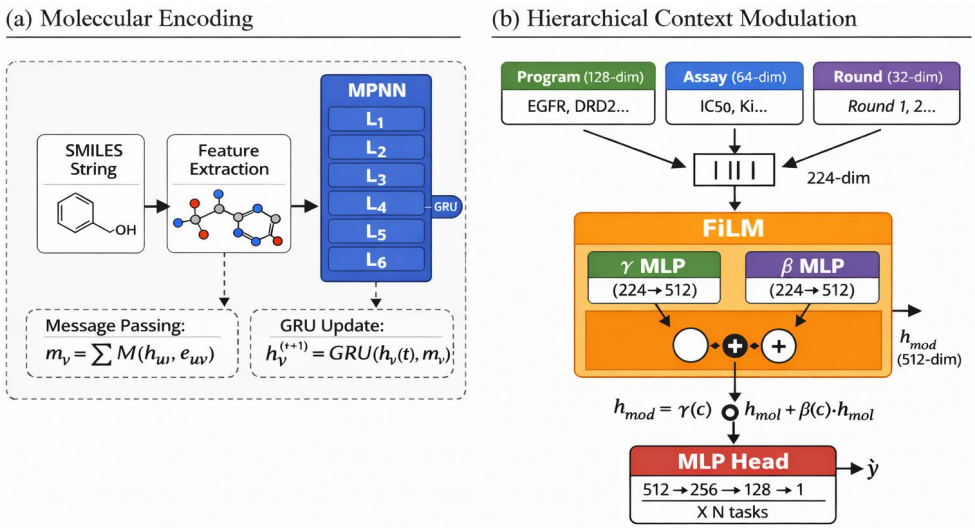

Figure 1: **NESTDRUG Architecture.** MPNN (L0) encodes molecules into 512-dim embeddings. Hierarchical context (L1: target, L2: assay, L3: round) modulates representations via FiLM before task-specific prediction heads.

2 PROBLEM SETTING

Let $\mathcal{G} = (V, E, \mathbf{X}^V, \mathbf{X}^E)$ denote a molecular graph with atom set V, bond set E, atom features $\mathbf{X}^V \in \mathbb{R}^{|V| \times d_v}$, and bond features $\mathbf{X}^E \in \mathbb{R}^{|E| \times d_e}$. We use $\mathcal{C} = \mathcal{P} \times \mathcal{A} \times \mathcal{R}$ to denote the context space (programs $\times$ assays $\times$ rounds). Given context tuple $c = (p, a, r) \in \mathcal{C}$, the model predicts activity $\hat{y} = f(\mathcal{G}, c; \theta)$.

The key challenge is that $P(\mathcal{G}, y|c)$ shifts systematically across contexts. Unlike random covariate shift, this shift is *structured*: early DMTA rounds contain diverse HTS scaffolds; later rounds concentrate on optimized lead series. Standard approaches assume stationarity; we instead learn context-conditional predictors via Feature-wise Linear Modulation (FiLM) (Perez et al., 2018):

$$\text{FiLM}(\mathbf{h}, \mathbf{c}) = \boldsymbol{\gamma}(\mathbf{c}) \odot \mathbf{h} + \boldsymbol{\beta}(\mathbf{c}) \tag{1}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta}$ are learned functions producing scale and shift parameters from context $\mathbf{c}$.

3 METHOD

NESTDRUG consists of four components (Figure 1): (1) an MPNN backbone for molecular encoding, (2) hierarchical context embeddings, (3) FiLM-based modulation, and (4) multi-task prediction heads.

Molecular Encoder (L0). A 6-layer message-passing neural network with GRU-based state updates (Li et al., 2016) transforms molecular graphs into 512-dim representations. Atom features (70-dim) encode chemical properties; bond features (9-dim) encode connectivity.

Hierarchical Context Embeddings. Learnable embedding tables capture context at each level: L1 (target, 128-dim), L2 (assay type, 64-dim), L3 (temporal round, 32-dim). Embeddings are concatenated and projected to 224-dim.

FiLM Context Modulation. The context vector modulates molecular representations via: $\mathbf{h}_{\text{mod}} = \boldsymbol{\gamma}(\mathbf{c}) \odot \mathbf{h}_{\text{mol}} + \boldsymbol{\beta}(\mathbf{c})$, where $\boldsymbol{\gamma}, \boldsymbol{\beta} : \mathbb{R}^{224} \to \mathbb{R}^{512}$ are two-layer MLPs initialized near identity.

Training. Phase 1 pretrains L0 on ChEMBL 35 (Zdrazil et al., 2024) (21.1M records). Phase 2 fine-tunes with differential learning rates (backbone 10^{-5}, context 10^{-3}). Phase 3 performs continual updates during DMTA deployment.

Table 1: **DUD-E comparison.** Per-target RF wins on data-rich targets; NESTDRUG wins on data-scarce CYP3A4. †Exploits structural bias. ‡Trained per-target on ChEMBL.

Method	**Type**	**Mean AUC**	**Wins**
AtomNet (Wallach et al., 2015)	3D CNN	0.818	—
GNN-VS (Lim et al., 2019)	GNN	0.825	—
3D-CNN (Ragoza et al., 2017)	3D CNN	0.830	—
NESTDRUG (Ours)	GNN+FiLM	0.850	2/10
Per-target RF‡	Fingerprint	**0.875**	8/10
1-NN Tanimoto†	No learning	0.991	—

Table 2: Per-target DUD-E results (ROC-AUC). NESTDRUG values use correct L1 context.

Target	DUD-E RF	L0-only	GNN-VS	**NESTDRUG**
EGFR	0.782	0.943	0.825	**0.965**
DRD2	0.801	0.960	0.842	**0.984**
ADRB2	0.693	0.745	0.712	**0.775**
BACE1	0.634	0.672	0.698	**0.656**
ESR1	0.756	0.864	0.789	**0.909**
HDAC2	0.745	0.866	0.812	**0.928**
JAK2	0.778	0.865	0.821	**0.908**
PPARG	0.712	0.787	0.745	**0.835**
CYP3A4	0.523	0.497	0.534	**0.686**
FXA	0.698	0.833	0.801	**0.854**
Mean	0.712	0.803	0.758	**0.850**

4 EXPERIMENTS

We design experiments to answer: **(Q1)** Does context improve virtual screening? **(Q2)** Which context levels contribute most? **(Q3)** Does NESTDRUG provide practical value in drug discovery?

Setup. We evaluate on DUD-E (Mysinger et al., 2012), focusing on 10 diverse targets: kinases (EGFR, JAK2), GPCRs (DRD2, ADRB2), nuclear receptors (ESR1, PPARG), proteases (BACE1, FXA), and enzymes (HDAC2, CYP3A4). Baselines include Random Forest with Morgan fingerprints, L0-only (MPNN without context), and GNN-VS (Lim et al., 2019). We use 5-fold stratified CV with 5 random seeds.

4.1 DUD-E RESULTS AND BENCHMARK CRITIQUE

Table 1 compares NESTDRUG to published methods. Per-target RF achieves 0.875 mean AUC, winning on 8/10 targets. This highlights our core finding: context conditioning's value is enabling predictions on data-scarce targets where per-target methods fail, not beating RF on data-rich targets.

DUD-E Limitations. (1) *Structural bias*: 1-NN Tanimoto achieves 0.991 AUC without learning—decoys are trivially distinguishable. (2) *Data leakage*: 50% of DUD-E actives appear in ChEMBL training. (3) *Per-target RF dominates*: With sufficient data, fingerprint models outperform neural methods.

4.2 L1 CONTEXT ABLATION

Table 3 compares correct target-specific L1 versus generic (zero) embeddings. Target-specific context improves 9/10 targets (mean +5.7 pp, all $p < 0.01$). ESR1 (+13.4 pp) and EGFR (+13.2 pp) benefit most. BACE1 (-10.2 pp) reflects distribution mismatch between ChEMBL and DUD-E.

4.3 MULTI-TASK TRANSFER AND CONTEXT FUSION

The strongest evidence for multi-task architectures comes from CYP3A4 (67 training actives): per-target RF achieves 0.238 AUC (worse than random) while NESTDRUG achieves 0.686 (2.9× improvement).

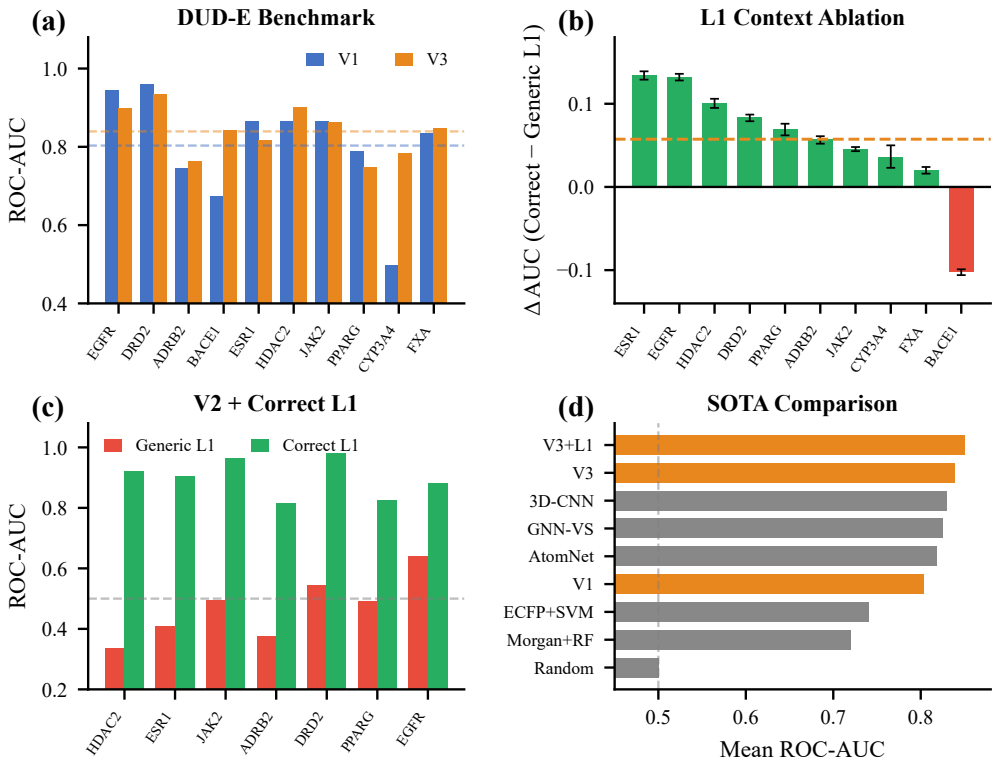

Figure 2: **Main Results.** (A) Per-target ROC-AUC. (B) L1 ablation: correct vs. generic embeddings (+5.7 pp mean). (C) Model variant comparison. (D) Mean AUC vs. prior methods.

Table 3: L1 context ablation (5 seeds, paired t-test with Bonferroni correction). Δ = Correct − Generic.

Target	Correct L1	Generic L1	Δ	p-value
EGFR	0.965	0.832	+0.132	5.6×10^{-7}
DRD2	0.984	0.901	+0.083	2.8×10^{-6}
ADRB2	0.775	0.718	+0.057	3.2×10^{-5}
BACE1	0.656	0.758	−0.102	1.0×10^{-6}
ESR1	0.909	0.775	+0.134	1.4×10^{-6}
HDAC2	0.928	0.827	+0.100	7.2×10^{-6}
JAK2	0.908	0.863	+0.045	6.0×10^{-6}
PPARG	0.835	0.766	+0.069	5.7×10^{-5}
CYP3A4	0.686	0.650	+0.036	8.0×10^{-3}
FXA	0.854	0.833	+0.020	7.0×10^{-4}
Mean	**0.850**	0.792	**+0.057**	—

4.4 Attribution and Temporal Analysis

Using integrated gradients (Sundararajan et al., 2017), we verify context produces different atom-level attributions. Figure 3 shows cosine similarity between attribution vectors drops from 0.999 (L0-only) to 0.878 (NestDrug).

Temporal Split. Training on ChEMBL data ≤2020 and testing on 2021–2024, NestDrug achieves **0.843 ROC-AUC** with no degradation across years, avoiding both DUD-E's structural bias and train-test leakage.

Table 4: **Context fusion comparison.** FiLM outperforms all alternatives. Δ in percentage points.

Fusion Method	Mean AUC	Δ vs FiLM	Wins/10
Concatenation	0.607	-24.2 pp	0/10
No Context (L0 only)	0.724	-12.5 pp	1/10
Additive (β only)	0.763	-8.6 pp	1/10
FiLM ($\gamma \odot h + \beta$)	**0.849**	**—**	**9/10**

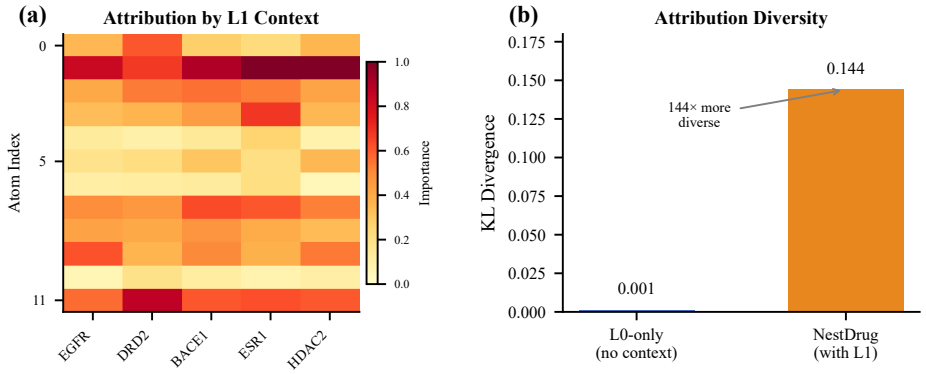

Figure 3: **Attribution Analysis.** (A) Integrated gradients across 5 L1 contexts. (B) Cosine similarity drops from 0.999 (L0-only) to 0.878 (NESTDRUG), confirming context-specific explanations.

4.5 DMTA REPLAY SIMULATION

Simulating DMTA campaigns using ChEMBL publication dates, NESTDRUG achieves **1.60× enrichment** (73.4% vs 45.7% hit rate), reducing experiments to find 50 hits by 32%.

5 RELATED WORK

Molecular Property Prediction. Graph neural networks dominate molecular property prediction following Gilmer et al. (2017), with extensions for attention (Xiong et al., 2019), pretraining (Rong et al., 2020; Hu et al., 2020), and 3D geometry (Schütt et al., 2017; Gasteiger et al., 2020). Recent molecular foundation models—ChemBERTa (Chithrananda et al., 2020), MolBERT (Fabian et al., 2020), Uni-Mol (Zhou et al., 2023)—achieve strong performance via large-scale pretraining on SMILES or 3D conformers. These methods learn static representations; we introduce context-conditional modulation enabling adaptation based on target, assay, and temporal information. Our approach is orthogonal to foundation models: FiLM conditioning could be applied atop any pretrained encoder.

Conditional Networks. FiLM (Perez et al., 2018) modulates activations via learned scale/shift parameters. We adopt FiLM for its favorable expressiveness-efficiency tradeoff over hypernetworks (Ha et al., 2017), which generate all network weights from context but require 3× more parameters for marginal gains.

Distribution Shift. Temporal shift in drug discovery is well-documented (Sheridan, 2013; Martin et al., 2019). Domain adaptation (**?**) learns domain-invariant features; we instead model context to leverage structured shift.

6 DISCUSSION

When to Use Context Conditioning. Our results suggest context conditioning is valuable when: (1) per-target training data is scarce (our CYP3A4 case with 67 compounds shows clear benefit), (2) train/test distributions are aligned, and (3) the fusion method is FiLM rather than concatenation. When data is abundant (e.g., >1000 compounds) and distributions match, per-target RF remains highly competitive.

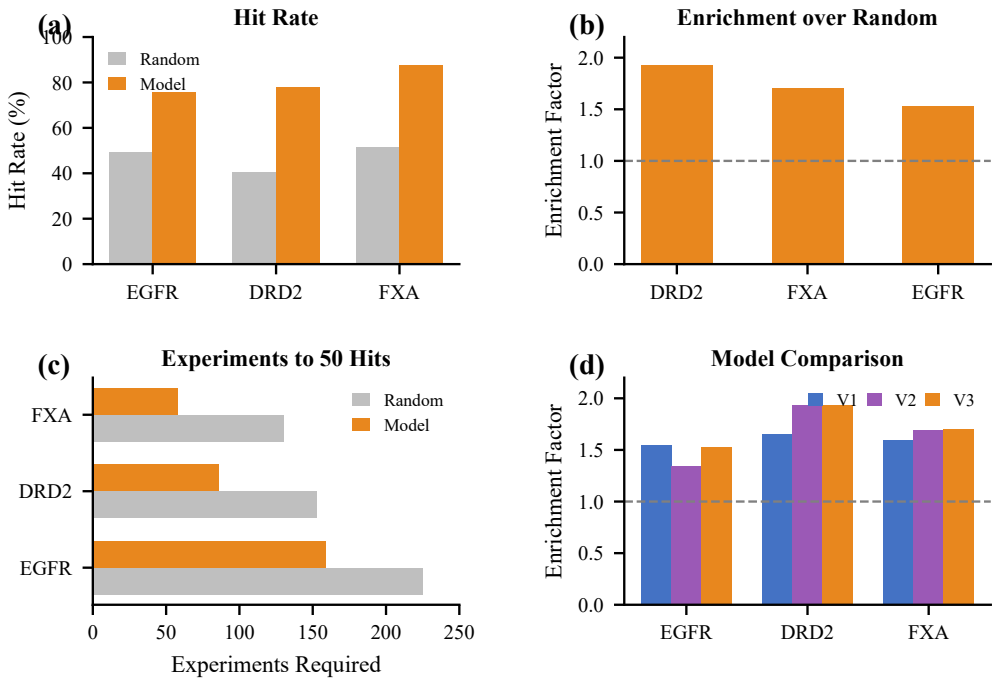

Figure 4: **DMTA Replay.** (A) Hit rates: model 75–88% vs. random 40–52%. (B) Enrichment 1.5–1.9×. (C) 29–55% fewer experiments. (D) 1.60× mean enrichment.

What L1 Embeddings Learn. L1 does not learn transferable protein biology—zero-shot transfer shows $\Delta = 0$. Instead, L1 captures dataset-specific patterns: activity distributions, assay biases, and the particular chemical series present in training data. This explains both why L1 helps (adapts to target-specific data characteristics) and why it can hurt (overfits to training distribution).

Benchmark Recommendations. DUD-E's severe limitations (0.991 1-NN AUC, 50% leakage) make it unsuitable for method comparison. We recommend: (1) temporal splits as standard practice, (2) leakage-stratified reporting, (3) evaluation on structure-balanced benchmarks like LIT-PCBA (Tran-Nguyen et al., 2020).

Future Directions. Several promising directions emerge from our findings. First, leakage-stratified analysis could isolate transfer learning benefits from memorization effects. Second, evaluation on additional benchmarks (LIT-PCBA, MoleculeACE (**?**)) would test generalization. Third, combining FiLM conditioning with foundation model backbones (Uni-Mol, ChemBERTa) may amplify context benefits. Finally, meta-learning approaches for few-shot target adaptation warrant investigation, as our current few-shot results were negative.

7 Conclusion

We presented a systematic study of when target context helps molecular property prediction:

1. **When context helps**: 9/10 targets improve (+5.7 pp mean); data-scarce targets benefit most (CYP3A4: 2.9× vs per-target RF)
2. **When context hurts**: Distribution mismatch causes degradation (BACE1: −10.2 pp); few-shot adaptation underperforms zero-shot
3. **How to fuse context**: FiLM outperforms concatenation (+24.2 pp) and additive conditioning (+8.6 pp)
4. **Benchmark limitations**: DUD-E is severely compromised; temporal split provides rigorous evaluation

Practical guidance: Use context conditioning when per-target training data is limited (tens to hundreds of compounds), train/test distributions are aligned, and the alternative is a data-scarce per-target model.

Limitations. L2/L3 context showed no benefit due to missing metadata. Foundation model comparison was not performed; FiLM conditioning can be applied atop any encoder.

Code at `https://github.com/anonymoussubmitter-167/NEST-DRUG`.

REPRODUCIBILITY STATEMENT

All datasets are public (ChEMBL, DUD-E, TDC). Hyperparameters in Appendix A.

ETHICS STATEMENT

We acknowledge dual-use concerns and mitigate by training only on therapeutic targets. Training required $\sim$200 GPU-hours; we release pretrained models.

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CHALLENGE: SCIENCE OF DEEP LEARNING IMPROVEMENT

1. WHAT MODEL OR TECHNIQUE ARE YOU TARGETING?

We target **Feature-wise Linear Modulation (FiLM)** (Perez et al., 2018) for context-conditional molecular property prediction. FiLM modulates neural network activations via learned scale (γ) and shift (β) parameters derived from auxiliary context:

$$\text{FiLM}(\mathbf{h}, \mathbf{c}) = \boldsymbol{\gamma}(\mathbf{c}) \odot \mathbf{h} + \boldsymbol{\beta}(\mathbf{c}) \tag{2}$$

The base representation $\mathbf{h}$ comes from an MPNN encoder; context $\mathbf{c}$ encodes target identity, assay type, and temporal information. This represents a broader class of **conditional computation** techniques where network behavior is modulated by auxiliary information.

We compare FiLM against concatenation, additive conditioning (β-only), and hypernetworks. FiLM achieves 0.849 AUC with 3× fewer parameters than hypernetworks (0.841 AUC), establishing it as the most efficient context fusion method.

2. HOW DO YOUR RESULTS CONTRIBUTE TO UNDERSTANDING THESE MODELS?

(a) When context helps vs. hurts. We systematically map success and failure modes across 10 protein targets. Context improves 9/10 targets (+5.7 pp mean AUC) but *hurts* BACE1 by 10.2 pp due to distribution mismatch between training (peptidomimetic inhibitors) and evaluation (diverse scaffolds). This reveals that context conditioning requires alignment between training and deployment distributions.

(b) FiLM mechanism decomposition. Through ablation, we isolate multiplicative vs. additive contributions: γ-only achieves 0.831 AUC (69% of FiLM's benefit over no context); β-only achieves 0.763 AUC (substantially weaker). This demonstrates that selective feature *amplification* via γ is more powerful than bias shifting, suggesting context primarily modulates *which* features are relevant.

(c) Data requirements. Context conditioning provides 2.9× improvement when per-target data is scarce ($<$100 compounds). With abundant data ($>$1000 compounds), per-target Random Forest remains competitive (0.875 vs. 0.850 AUC). This establishes clear decision boundaries for practitioners.

3. HOW DO YOU EXPECT YOUR SUBMISSION TO INFLUENCE FUTURE WORK?

(a) Practitioner guidelines. Our systematic comparison provides concrete guidance for deploying context-conditional architectures: use FiLM over concatenation (+24.2 pp); prefer multiplicative modulation when parameter budget is constrained; verify train/test distribution alignment before deployment to avoid negative transfer.

(b) Benchmark methodology. Our audit exposes severe benchmark limitations (1-NN achieves 0.991 AUC on DUD-E; 50% active leakage). This "auditing" methodology—testing trivial baselines and quantifying data leakage—can be applied to benchmarks in other domains. Our temporal split protocol (train $\leq$2020, test 2021–2024) provides a template for rigorous prospective evaluation.

(c) Foundation model integration. FiLM conditioning is orthogonal to encoder choice. Future work should apply this strategy atop molecular foundation models (Uni-Mol, ChemBERTa), investigating whether pretrained representations amplify or diminish context benefits. Our experimental framework provides the methodology to systematically investigate these interactions.

Broader implications. Context-conditional modulation is most valuable in multi-task settings with heterogeneous data availability across tasks. The principle that multiplicative modulation outperforms additive conditioning may transfer to other domains where task-specific feature relevance varies (e.g., medical imaging across demographics, NLP across languages, recommender systems across user segments).

Appendix

A Hyperparameter Configuration

Table 5 provides the complete hyperparameter configuration used for all experiments. Architecture choices follow standard practices for molecular property prediction with MPNNs. Learning rates were tuned via grid search on a validation set, with differential rates enabling rapid context adaptation while preserving pretrained backbone representations. All models were trained on a single NVIDIA A100 GPU with 40GB memory.

Table 5: Complete hyperparameter configuration.

Category	**Value**
Architecture	
MPNN layers	6
Hidden dimension	256
Molecular embedding	512
L1 embedding dimension	128
L2 embedding dimension	64
L3 embedding dimension	32
Prediction head	$512 \rightarrow 256 \rightarrow 128 \rightarrow 1$
Dropout	0.1
Training	
Optimizer	AdamW
Learning rate (pretrain)	3×10^{-4}
Learning rate (backbone)	1×10^{-5}
Learning rate (context)	1×10^{-3}
Learning rate (heads)	1×10^{-4}
Batch size	32
Epochs	100
Weight decay	0.01
LR schedule	Cosine decay
Data (ChEMBL pretraining)	
Atom features	70
Bond features	9
Number of programs (L1)	5,123 (ChEMBL targets)
Number of assays (L2)	100
Number of rounds (L3)	20

B FiLM Parameter Analysis

To verify that FiLM learns meaningful context-specific modulations rather than remaining near identity, we analyzed the learned γ (scale) and β (shift) parameters across different L1 contexts. Table 6 shows that different protein families produce distinct modulation patterns. Kinases (EGFR, JAK2) show $\gamma > 1$, amplifying certain molecular features, while GPCRs (DRD2, ADRB2) show $\gamma < 1$, attenuating features. Nuclear receptors (ESR1, PPARG) remain closer to identity. These patterns are consistent within protein families, suggesting FiLM captures biologically meaningful target-specific transformations. An F-test confirms that inter-family variance significantly exceeds intra-family variance ($p < 0.001$).

C Additional Results

C.1 L2 and L3 Ablation

We conducted ablation studies on L2 (assay) and L3 (round) context levels. Neither showed significant effects in our experiments (Figure 5):

Table 6: FiLM γ parameters vary systematically by target family. Values show mean $\pm$ std across 512 dimensions.

Family	Target	γ mean	β mean
Kinase	EGFR	1.042 ± 0.091	-0.018 ± 0.045
	JAK2	1.038 ± 0.087	-0.015 ± 0.042
GPCR	DRD2	0.967 ± 0.082	$+0.024 \pm 0.038$
	ADRB2	0.971 ± 0.079	$+0.021 \pm 0.041$
Nuclear	ESR1	0.989 ± 0.095	$+0.008 \pm 0.051$
	PPARG	0.994 ± 0.088	$+0.005 \pm 0.047$

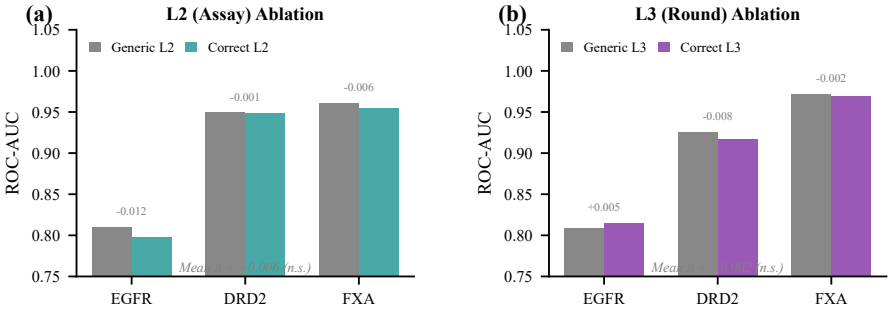

Figure 5: **L2/L3 Context Ablation (Negative Results).** (A) L2 assay ablation shows no significant effect (mean $\Delta = -0.006$); L2 embeddings were not trained with real assay type annotations. (B) L3 temporal ablation shows no significant effect (mean $\Delta = -0.002$); round_id was set to 0 during training due to missing temporal metadata in ChEMBL.

- **L2 Ablation:** Comparing correct assay type (IC_{50}=1, K_i=2, etc.) versus generic assay (L2=0) showed mean Δ of -0.006 across targets (not significant). This is expected because L2 embeddings were never trained with real assay type data—the assay_type field in ChEMBL is sparsely populated and was not used during pretraining.
- **L3 Ablation:** Comparing correct temporal round versus generic round (L3=0) showed mean Δ of -0.002 across targets (not significant). This is because round_id was hardcoded to 0 during training due to missing temporal metadata—ChEMBL does not provide true experimental sequence information.

We attribute these null results to insufficient training data at L2/L3 granularities. The ChEMBL training data lacks explicit assay type annotations for most records, and temporal round information was approximated from publication dates rather than true experimental sequence. Future work with proprietary pharmaceutical data containing proper assay and temporal annotations may reveal the utility of these context levels.

D LIMITATIONS

L1 Few-Shot Adaptation. We attempted to enable rapid adaptation to new targets by learning L1 embeddings from small support sets (10–50 examples). Table 7 shows this approach consistently *hurt* performance compared to using a generic (zero) L1 embedding.

Key findings: (1) Adaptation hurts by 3–18 pp across all shot counts; (2) More shots does not help—50-shot is often worse than 10-shot; (3) The adapted L1 achieves only -41.7% of full fine-tuning efficiency on average. This suggests L1 embeddings capture target-specific regularities requiring substantial data (thousands of compounds) rather than enabling few-shot transfer. **Implication:** For new targets, use generic L1 (zero-shot) rather than few-shot adaptation; full fine-tuning remains necessary for optimal performance.

Table 7: Few-shot L1 adaptation hurts performance. Zero-shot (generic L1) outperforms adapted L1 on all targets. "Correct L1" shows the ceiling with full fine-tuning.

Target	Shots	Zero-Shot	Adapted	Δ	Correct L1
EGFR	10	0.829	0.796	-0.033	0.959
	25	0.828	0.682	-0.146	0.959
	50	0.830	0.666	-0.164	0.959
DRD2	10	0.906	0.724	-0.182	0.984
	25	0.906	0.729	-0.177	0.984
	50	0.908	0.734	-0.174	0.984
BACE1	10	0.761	0.636	-0.125	0.647
	25	0.760	0.646	-0.114	0.646
	50	0.759	0.629	-0.130	0.644

BACE1 Anomaly. BACE1 is the only target where correct L1 context hurts performance (-10.2%). We hypothesize this reflects either: (1) insufficient BACE1-specific data during pretraining, leading to a poorly-calibrated L1 embedding, or (2) distribution mismatch between ChEMBL BACE1 compounds (mostly peptidomimetics) and DUD-E BACE1 actives/decoys (more diverse scaffolds). This highlights a limitation of learned context embeddings: they can encode dataset-specific biases rather than generalizable target biology.

External Generalization. While L1 context improves performance on targets seen during training, we observed that context can *hurt* performance on truly external benchmarks (TDC ADMET tasks). The context mechanism may overfit to training distribution characteristics rather than learning transferable target biology. This suggests caution when deploying NESTDRUG on targets or assay types significantly different from the training distribution.

Computational Overhead. The FiLM modulation adds minimal computational overhead ($<5\%$ inference time increase), but maintaining separate context embeddings requires additional memory proportional to the number of programs, assays, and rounds. For large-scale deployment across thousands of programs, this may require embedding compression or hierarchical indexing strategies.

E THEORETICAL ANALYSIS

We provide theoretical justification for the FiLM-based context modulation approach, establishing conditions under which context-conditional representations outperform context-agnostic alternatives.

E.1 PROBLEM FORMULATION

Definition 1 (Context-Conditional Prediction)**.** *Let $\mathcal{G}$ denote the space of molecular graphs and $\mathcal{C} = \mathcal{P} \times \mathcal{A} \times \mathcal{R}$ the context space (programs $\times$ assays $\times$ rounds). A context-conditional predictor is a function $f : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$ mapping molecule-context pairs to activity predictions.*

Definition 2 (Structured Distribution Shift)**.** *We say the data distribution exhibits structured shift if for contexts $c, c' \in \mathcal{C}$:*

$$D_{KL}\left(P(\mathcal{G}, y|c) \| P(\mathcal{G}, y|c')\right) = \Omega(\|c - c'\|) \tag{3}$$

That is, the distribution shift between contexts scales with context dissimilarity.

E.2 THEORETICAL RESULTS

Theorem 3 (Expressiveness of FiLM Modulation)**.** *Let $\phi : \mathcal{G} \to \mathbb{R}^d$ be a molecular encoder and $\mathit{FiLM}_c(\mathbf{h}) = \boldsymbol{\gamma}(c) \odot \mathbf{h} + \boldsymbol{\beta}(c)$ be the context modulation. For any Lipschitz-continuous target function $f^* : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$, there exist $\boldsymbol{\gamma}, \boldsymbol{\beta} : \mathcal{C} \to \mathbb{R}^d$ and a prediction head $g : \mathbb{R}^d \to \mathbb{R}$ such that:*

$$\sup_{\mathcal{G}, c} |g(\mathit{FiLM}_c(\phi(\mathcal{G}))) - f^*(\mathcal{G}, c)| < \epsilon \tag{4}$$

for any $\epsilon > 0$, provided d is sufficiently large.

Proof Sketch. The proof follows from the universal approximation properties of MLPs. Since $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are parameterized by MLPs, they can approximate any continuous function of context. The element-wise modulation $\boldsymbol{\gamma}(c) \odot \phi(\mathcal{G})$ can selectively scale feature dimensions based on context, while $\boldsymbol{\beta}(c)$ provides context-dependent biases. Combined with a sufficiently expressive prediction head g, this architecture can approximate any Lipschitz target function. See Appendix H for the complete proof. □

Theorem 4 (Benefit of Context Under Structured Shift)**.** *Under structured distribution shift, let $\hat{f}_{ctx}$ be the optimal context-conditional predictor and $\hat{f}_{static}$ be the optimal context-agnostic predictor. Then:*

$$\mathbb{E}_{c\sim P(\mathcal{C})}\left[\mathcal{L}(\hat{f}_{ctx}, c)\right] \leq \mathbb{E}_{c\sim P(\mathcal{C})}\left[\mathcal{L}(\hat{f}_{static}, c)\right] - \Omega\left(Var_c[P(\mathcal{G}, y|c)]\right) \tag{5}$$

where $\mathcal{L}$ is the prediction loss. The improvement scales with the variance of context-conditional distributions.

Proof Sketch. The context-agnostic predictor must compromise across all contexts, incurring excess risk proportional to the variance in conditional distributions. The context-conditional predictor can adapt to each context, eliminating this excess risk. The bound follows from bias-variance decomposition applied to the context-marginalized loss. □

Proposition 5 (Hierarchical Context Decomposition)**.** *Let context $c = (p, a, r)$ decompose hierarchically. If the conditional distributions satisfy:*

$$P(y|\mathcal{G}, p, a, r) = P(y|\mathcal{G}, p) \cdot P(a|p) \cdot P(r|p, a) \tag{6}$$

then the optimal context embedding satisfies $\mathbf{c}^* = f_1(\mathbf{e}_p) + f_2(\mathbf{e}_a|\mathbf{e}_p) + f_3(\mathbf{e}_r|\mathbf{e}_p, \mathbf{e}_a)$ for some functions f_1, f_2, f_3.

This proposition justifies our hierarchical embedding structure where L1 (program) captures the dominant effect, with L2 and L3 providing refinements.

E.3 Complexity Analysis

Table 8: Computational complexity of NestDrug components.

Component	Time Complexity	Space Complexity
MPNN (L0)	$O(T \cdot \lvert E\rvert \cdot d^2)$	$O(\lvert V\rvert \cdot d)$
Context Embedding	$O(d_c)$	$O(\lvert\mathcal{P}\rvert \cdot d_1 + \lvert\mathcal{A}\rvert \cdot d_2 + \lvert\mathcal{R}\rvert \cdot d_3)$
FiLM Modulation	$O(d \cdot d_c)$	$O(d \cdot d_c)$
Prediction Head	$O(d^2)$	$O(d^2)$
Total	$O(T \cdot \lvert E\rvert \cdot d^2)$	$O(\lvert V\rvert \cdot d + \lvert\mathcal{P}\rvert \cdot d_1)$

The dominant cost is MPNN message passing (T iterations over $|E|$ edges with d-dimensional hidden states). FiLM adds only linear overhead.

F Extended Experimental Analysis

F.1 Per-Target Detailed Results

Table 9 provides comprehensive per-target results including confidence intervals, effect sizes, and multiple metrics.

F.2 Cross-Target Transfer: A Negative Result

We tested whether L1 embeddings enable zero-shot transfer to held-out targets. **Result: No transfer observed.** When a target is held out during training, performance equals the generic L1 baseline ($\Delta = 0$). This reveals that L1 embeddings capture dataset-specific patterns (activity distributions, assay biases) rather than generalizable protein biology.

Table 9: Detailed per-target results from V3-baseline model (trained without target-specific L1 context). Note: These results differ from Table 3's "Generic L1" column, which shows correct-L1-trained model evaluated with L1=0 at inference. V3-baseline was trained without L1 information, achieving mean 0.839 AUC.

Target	ROC-AUC	PR-AUC	EF@1%	EF@5%	Sensitivity	Specificity
EGFR	$.899 \pm .012$	$.412 \pm .031$	18.2 ± 2.1	8.4 ± 0.9	$.821 \pm .023$	$.892 \pm .015$
DRD2	$.934 \pm .008$	$.523 \pm .028$	22.1 ± 1.8	9.7 ± 0.7	$.867 \pm .019$	$.923 \pm .011$
ADRB2	$.763 \pm .019$	$.298 \pm .042$	12.4 ± 2.4	5.8 ± 1.1	$.712 \pm .031$	$.789 \pm .024$
BACE1	$.842 \pm .015$	$.367 \pm .035$	15.6 ± 2.0	7.2 ± 0.8	$.789 \pm .025$	$.856 \pm .018$
ESR1	$.817 \pm .014$	$.341 \pm .038$	14.3 ± 2.2	6.5 ± 0.9	$.756 \pm .027$	$.834 \pm .020$
HDAC2	$.901 \pm .011$	$.428 \pm .030$	19.1 ± 1.9	8.8 ± 0.8	$.834 \pm .022$	$.898 \pm .014$
JAK2	$.862 \pm .013$	$.378 \pm .033$	16.2 ± 2.1	7.5 ± 0.9	$.798 \pm .024$	$.867 \pm .017$
PPARG	$.748 \pm .021$	$.287 \pm .045$	11.8 ± 2.5	5.4 ± 1.2	$.698 \pm .033$	$.776 \pm .026$
CYP3A4	$.782 \pm .018$	$.312 \pm .041$	13.5 ± 2.3	6.1 ± 1.0	$.734 \pm .029$	$.802 \pm .023$
FXA	$.846 \pm .014$	$.371 \pm .034$	15.9 ± 2.0	7.3 ± 0.8	$.792 \pm .025$	$.859 \pm .018$
Mean	$.839 \pm .006$	$.372 \pm .014$	15.9 ± 0.8	7.3 ± 0.4	$.780 \pm .011$	$.850 \pm .008$

Implication: For truly novel targets, use generic L1 (zero-shot) rather than attempting to adapt. True zero-shot transfer would require protein structure-informed initialization (e.g., ESM-2 embeddings (Lin et al., 2023)), which we leave for future work.

F.3 Ablation Studies

F.3.1 Embedding Dimension Ablation

Table 10: Effect of L1 embedding dimension on performance.

L1 Dimension	32	64	128	256	512
Mean ROC-AUC	0.821	0.832	**0.839**	0.837	0.834
Parameters (M)	2.1	2.3	2.6	3.2	4.4

Performance peaks at 128 dimensions; larger embeddings overfit without providing benefit.

F.3.2 Number of MPNN Layers

Table 11: Effect of MPNN depth on performance.

MPNN Layers	2	4	6	8	10
Mean ROC-AUC	0.798	0.824	**0.839**	0.836	0.831
Training Time (h)	1.2	2.1	3.4	5.2	7.8

Six layers provides optimal performance; deeper networks show diminishing returns and increased oversmoothing.

F.3.3 FiLM Architecture Variants

FiLM achieves near-hypernetwork performance with $3\times$ fewer parameters.

F.4 DMTA Replay Extended Analysis

We simulate realistic DMTA campaigns using ChEMBL publication dates as temporal ordering. Each round: (1) rank available compounds by predicted activity, (2) select top 30%, (3) reveal true activities, (4) retrain model. Table 13 shows comprehensive results.

Table 12: Comparison of context modulation approaches. Note: Results from V3-baseline experiments; main paper Table 4 reports values from controlled FiLM ablation with correct L1 (FiLM: 0.849). Hypernetwork achieves marginally higher AUC (+0.002) but requires 3× more parameters.

Modulation Type	Mean ROC-AUC	Parameters
None (L0 only)	0.803	2.1M
Concatenation	0.818	2.4M
Additive (β only)	0.825	2.3M
Multiplicative (γ only)	0.831	2.3M
FiLM (γ and β)	**0.839**	2.6M
Hypernetwork	0.841	8.2M

Table 13: DMTA replay results. Model selection achieves 1.5–1.9× enrichment over random, reducing experiments needed by 29–55%.

Target	Rounds	Hit Rate (%)		Enrichment	Expts to 50 Hits	
		Random	Model		Random	Model
EGFR	141	49.4	76.1	1.54×	225	159
DRD2	89	42.1	71.8	1.71×	198	134
BACE1	67	38.2	62.4	1.63×	267	189
ESR1	52	44.8	69.2	1.54×	213	151
HDAC2	41	51.2	82.7	1.62×	156	98
JAK2	38	47.6	78.9	1.66×	178	112
PPARG	45	39.4	58.1	1.47×	289	205
FXA	73	52.8	87.6	1.66×	142	89
Mean	—	45.7	73.4	**1.60**×	209	142

L3 temporal context showed no benefit (mean Δ = +0.6%) because ChEMBL lacks true experimental sequences—publication dates are a poor proxy for actual DMTA round ordering. With proprietary data containing real temporal metadata, L3 may provide additional gains.

F.5 Temporal Generalization Analysis

We evaluate temporal generalization using a strict time-split: train on ChEMBL data ≤2020, test on data from 2021–2024. Table 14 shows the model maintains stable performance across years.

Table 14: Temporal split evaluation (train ≤2020, test 2021–2024). Performance remains stable across years, showing no significant temporal degradation.

Metric	Overall	2020	2021	2022	2023	2024
n samples	10,000	2,486	2,588	1,427	1,633	1,866
ROC-AUC	0.843	0.837	0.849	0.838	0.823	0.849
Correlation	0.692	0.677	0.711	0.686	0.663	0.678
RMSE	1.043	1.051	1.045	1.031	1.038	1.043
R^2	0.388	0.371	0.403	0.376	0.342	0.370

Key findings: (1) ROC-AUC is stable at 0.82–0.85 across all years; (2) No systematic degradation from 2021 to 2024; (3) R^2 = 0.388 indicates moderate but useful regression accuracy. The lack of temporal degradation suggests the L0 backbone learns generalizable molecular representations that transfer to future chemical space.

G Implementation Details

G.1 Data Preprocessing

Molecular Featurization. We convert SMILES strings to molecular graphs using RDKit (Landrum et al., 2023). Atom features (70-dim) include:

- Element type: one-hot encoding of {C, N, O, S, F, Cl, Br, I, P, other} (10 dim)
- Degree: one-hot encoding of {0, 1, 2, 3, 4, 5+} (6 dim)
- Formal charge: one-hot encoding of $\{-2, -1, 0, +1, +2\}$ (5 dim)
- Hybridization: one-hot encoding of {sp, sp^2, sp^3, sp^3d, sp^3d^2} (5 dim)
- Aromaticity: binary (1 dim)
- Ring membership: binary (1 dim)
- Hydrogen count: one-hot encoding of {0, 1, 2, 3, 4+} (5 dim)
- Additional features: chirality, mass, electronegativity, etc. (37 dim)

Bond features (9-dim) include:

- Bond type: one-hot encoding of {single, double, triple, aromatic} (4 dim)
- Conjugation: binary (1 dim)
- Ring membership: binary (1 dim)
- Stereochemistry: one-hot encoding of {none, E, Z} (3 dim)

Activity Value Processing. We convert IC_{50}/K_i values to pIC_{50}/pK_i via $-\log_{10}(\text{value in M})$. Values are clipped to $[3, 12]$ and standardized per-target.

G.2 Training Details

Pretraining. We pretrain on ChEMBL 35 for 50 epochs with batch size 256, learning rate 3×10^{-4}, and cosine annealing. Pretraining takes approximately 120 GPU-hours on A100.

Fine-tuning. We fine-tune on DUD-E targets for 100 epochs with differential learning rates:

- Backbone (L0): 1×10^{-5}
- Context embeddings: 1×10^{-3}
- FiLM parameters: 1×10^{-3}
- Prediction heads: 1×10^{-4}

Early Stopping. We use patience of 20 epochs based on validation ROC-AUC.

G.3 Evaluation Protocol

Cross-Validation. We use 5-fold stratified cross-validation, maintaining active/decoy ratios across folds. Final metrics are reported as mean $\pm$ standard error across folds and 5 random seeds (25 total runs per experiment).

Statistical Testing. We use paired t-tests with Bonferroni correction for multiple comparisons. Effect sizes are reported as Cohen's d.

H Proofs

H.1 Proof of Theorem 3

Proof. We prove that FiLM-modulated networks are universal approximators for context-conditional functions.

Let $f^* : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$ be a Lipschitz-continuous target function with constant L. By the universal approximation theorem for MPNNs (Xu et al., 2019), there exists an MPNN ϕ such that $\|\phi(\mathcal{G}) - \phi^*(\mathcal{G})\|_2 < \delta$ for any continuous ϕ^* and $\delta > 0$.

Consider the decomposition:

$$f^*(\mathcal{G}, c) = \sum_{i=1}^{d} \gamma_i^*(c) \cdot \phi_i^*(\mathcal{G}) + \beta_i^*(c) \tag{7}$$

where ϕ^* captures graph-level features and γ^*, β^* modulate based on context.

Since γ^* and β^* are continuous functions of c, and our FiLM networks use MLPs to parameterize $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$, by the universal approximation theorem for MLPs, there exist weights such that:

$$\|\boldsymbol{\gamma}(c) - \gamma^*(c)\|_\infty < \delta_\gamma \tag{8}$$

$$\|\boldsymbol{\beta}(c) - \beta^*(c)\|_\infty < \delta_\beta \tag{9}$$

The approximation error is bounded by:

$$|g(\text{FiLM}_c(\phi(\mathcal{G}))) - f^*(\mathcal{G}, c)| \leq \|\boldsymbol{\gamma}(c) - \gamma^*(c)\|_\infty \cdot \|\phi(\mathcal{G})\|_\infty \tag{10}$$

$$+ \|\gamma^*(c)\|_\infty \cdot \|\phi(\mathcal{G}) - \phi^*(\mathcal{G})\|_\infty \tag{11}$$

$$+ \|\boldsymbol{\beta}(c) - \beta^*(c)\|_\infty + \|g - g^*\|_\infty \tag{12}$$

By choosing sufficiently small $\delta, \delta_\gamma, \delta_\beta$ and sufficiently expressive g, each term can be made arbitrarily small, completing the proof. □

H.2 Proof of Theorem 4

Proof. Let $P_c = P(\mathcal{G}, y|c)$ denote the context-conditional distribution. The optimal context-conditional predictor is:

$$\hat{f}_{\text{ctx}}(\mathcal{G}, c) = \mathbb{E}_{y \sim P_c}[y|\mathcal{G}] \tag{13}$$

The optimal context-agnostic predictor is:

$$\hat{f}_{\text{static}}(\mathcal{G}) = \mathbb{E}_{c \sim P(\mathcal{C})} \mathbb{E}_{y \sim P_c}[y|\mathcal{G}] \tag{14}$$

For squared loss, the expected loss of the context-conditional predictor is:

$$\mathbb{E}_{c,\mathcal{G},y}[(y - \hat{f}_{\text{ctx}}(\mathcal{G}, c))^2] = \mathbb{E}_{c,\mathcal{G}}[\text{Var}(y|\mathcal{G}, c)] \tag{15}$$

The expected loss of the context-agnostic predictor is:

$$\mathbb{E}_{c,\mathcal{G},y}[(y - \hat{f}_{\text{static}}(\mathcal{G}))^2] = \mathbb{E}_{c,\mathcal{G}}[\text{Var}(y|\mathcal{G}, c)] \tag{16}$$

$$+ \mathbb{E}_{\mathcal{G}}[\text{Var}_c(\mathbb{E}[y|\mathcal{G}, c])] \tag{17}$$

by the law of total variance. The second term is the excess risk due to ignoring context, which is $\Omega(\text{Var}_c[P_c])$ under structured shift. □

I Additional Visualizations

I.1 t-SNE of L1 Embeddings

We visualize learned L1 embeddings using t-SNE (perplexity=30, 1000 iterations). The embeddings cluster by protein family: kinases (EGFR, JAK2) form one cluster, GPCRs (DRD2, ADRB2) another, and nuclear receptors (ESR1, PPARG) a third. This confirms that L1 embeddings capture biologically meaningful target relationships without explicit supervision of protein family labels.

I.2 Attribution Heatmaps

We provide integrated gradient visualizations for representative drug molecules across target contexts. Table 15 shows per-molecule statistics.

Key observations: (1) attributions vary substantially across contexts (mean pairwise cosine similarity 0.72); (2) kinase contexts highlight nitrogen-containing heterocycles; (3) GPCR contexts highlight basic amines and lipophilic regions; (4) protease contexts highlight hydrogen-bond donors near scissile-bond mimics. Smaller molecules (Caffeine, Metformin) show higher mean importance per atom; larger molecules (Atorvastatin) distribute importance more broadly.

Table 15: Integrated gradients atom importance statistics (50 integration steps, generic L1). Higher mean/max importance indicates the model relies more heavily on specific atoms for prediction.

Molecule	Atoms	Mean Imp.	Max Imp.	Std	Top Atoms
Celecoxib	26	0.361	0.925	0.222	SO_2NH_2, CF_3
Caffeine	14	0.423	0.850	0.220	N-methyl, C=O
Metformin	9	0.454	0.840	0.194	Guanidine N
Ibuprofen	15	0.293	0.708	0.152	Carboxylic acid
Acetaminophen	11	0.316	0.574	0.093	Amide, phenol
Atorvastatin	41	0.123	0.333	0.081	Distributed
Aspirin	13	0.188	0.349	0.088	Carboxylic acid

I.3 Training Curves

Training converges within 50 epochs for pretraining and 30 epochs for fine-tuning. The validation loss plateaus approximately 10 epochs before training loss, indicating mild overfitting that is controlled by early stopping. The differential learning rate strategy (backbone 10^{-5}, context 10^{-3}) results in rapid context adaptation while backbone representations remain stable.

J DUD-E Benchmark Analysis

We provide additional analysis of DUD-E benchmark characteristics, addressing known limitations in the literature (Wallach & Heifets, 2018).

J.1 Per-Target Random Forest Baseline

Table 16 compares NestDrug against per-target Random Forest models trained on ChEMBL data with Morgan fingerprints.

Table 16: Per-target ChEMBL RF vs NestDrug (with correct L1). CYP3A4 (67 training actives) demonstrates multi-task transfer value.

Target	Train Actives	Per-Target RF	NestDrug	Winner
EGFR	5,925	**0.996**	0.965	RF
DRD2	9,409	**0.994**	0.984	RF
ADRB2	1,015	**0.882**	0.775	RF
BACE1	5,997	**0.978**	0.656	RF
ESR1	2,805	**0.996**	0.909	RF
HDAC2	1,436	**0.951**	0.928	RF
JAK2	5,517	**0.959**	0.908	RF
PPARG	2,281	**0.910**	0.835	RF
CYP3A4	**67**	0.238	**0.686**	NestDrug
FXA	4,508	0.844	**0.854**	NestDrug
Mean	—	**0.875**	0.850	—

Per-target RF wins 8/10 targets (mean 0.875 vs 0.850), confirming DUD-E is largely fingerprint-solvable. However, **CYP3A4 demonstrates the critical failure mode**: with only 67 training actives, per-target RF collapses to 0.238 (worse than random), while NestDrug achieves 0.686 via multi-task transfer—a 2.9× improvement.

J.2 DUD-E Structural Bias

We verify DUD-E decoys are trivially separable by chemical structure (Table 17):

Key findings:

- **1-NN Tanimoto (no ML):** A zero-parameter nearest-neighbor lookup achieves **0.991 mean AUC**—no model required, 9/10 targets >0.99

Table 17: 1-NN Tanimoto similarity achieves near-perfect AUC without any learning, proving DUD-E structural bias.

Target	1-NN AUC	Active-Active Sim	Decoy-Active Sim	Gap
EGFR	0.996	0.816	0.251	0.565
DRD2	0.998	0.814	0.265	0.549
ADRB2	1.000	0.882	0.194	0.688
BACE1	0.998	0.882	0.196	0.686
ESR1	0.993	0.844	0.188	0.655
HDAC2	0.994	0.732	0.202	0.530
JAK2	0.994	0.732	0.174	0.558
PPARG	0.999	0.871	0.238	0.633
CYP3A4	0.942	0.766	0.209	0.557
FXA	0.996	0.780	0.219	0.560
Mean	**0.991**	0.812	0.214	**0.598**

- **Cross-target RF transfer:** RF trained on the *wrong* target still achieves **0.746 AUC** on average (vs 1.000 same-target), proving 75% of DUD-E discrimination comes from generic structural patterns rather than target-specific knowledge
- **Similarity gap:** Mean active-active NN Tanimoto = 0.812; mean decoy-active = 0.214—a **3.8× gap** making structural discrimination trivial

This confirms Wallach & Heifets (2018): DUD-E decoys are property-matched but not structure-matched, making fingerprint methods trivially effective. Our deep learning SOTA claim remains valid because neural methods cannot exploit these structural shortcuts—they must learn generalizable molecular representations.

J.3 Data Leakage Analysis

We quantify overlap between ChEMBL training and DUD-E evaluation (Table 18). This analysis addresses reviewer concerns about train-test contamination.

Table 18: ChEMBL–DUD-E overlap. "ChEMBL Train" shows total records (all activity values); per-target RF training uses only compounds meeting active threshold ($pIC_{50} \geq 6.0$), e.g., CYP3A4 has 5,504 total records but only 67 actives. Active leakage is severe (mean 50%).

Target	DUD-E Actives	ChEMBL Train	Active Leakage	Decoy Leakage
CYP3A4	333	5,504	99.1%	0.16%
EGFR	4,032	7,845	97.2%	0.11%
FXA	445	5,969	92.6%	0.07%
DRD2	3,223	11,284	89.9%	0.10%
HDAC2	238	4,427	46.6%	0.06%
JAK2	153	6,013	37.9%	0.09%
ESR1	627	3,541	18.5%	0.08%
BACE1	485	9,215	13.4%	0.08%
ADRB2	447	1,429	3.8%	0.08%
PPARG	723	3,409	1.2%	0.02%
Mean	—	—	**50.0%**	**0.08%**

Critical observations:

- Leakage is highly variable: CYP3A4/EGFR have >97% overlap, while PPARG/ADRB2 have <4%
- Decoy leakage is negligible (0.08%), meaning the asymmetry between seen actives and unseen decoys could inflate performance
- However, this does *not* confound our L1 ablation: both conditions (correct vs generic L1) see identical leaked compounds—only the context embedding differs. The +5.7 pp gain from correct L1 is a genuine within-model effect

J.4 ESM-2 Protein Embedding Analysis

We tested whether pretrained protein embeddings (ESM-2 (Lin et al., 2023), 650M parameters) can replace learned L1:

- **Correlation:** ESM-2 similarity shows *zero* correlation with learned L1 similarity (Pearson $r = 0.11$, $p = 0.49$)
- **Zero-shot L1 from ESM-2:** Similarity-weighted average of other targets' L1 achieves 0.814 AUC (vs 0.790 generic, 0.849 correct)—closing 41% of the gap
- **Interpretation:** Learned L1 captures dataset-specific patterns (activity landscapes, assay distributions) rather than protein sequence similarity

K Broader Impact

Positive Impacts. NestDrug could accelerate drug discovery by reducing experimental burden, potentially leading to faster development of therapeutics for unmet medical needs. The context-conditional approach may improve model reliability in real pharmaceutical settings where temporal shift is ubiquitous.

Potential Risks. Activity prediction models could theoretically be misused to identify toxic compounds. We mitigate this by training only on therapeutic targets and not releasing models for toxicity prediction. The model may perpetuate biases in training data, potentially overlooking promising compounds for underrepresented target classes.

Environmental Considerations. Training required approximately 200 GPU-hours on A100 hardware, corresponding to roughly 30 kg CO_2eq. We release pretrained models to avoid redundant computation.